

Chapter 4

Conjugate Gradient Method

In the preceding chapter we have discussed the steepest descent method and the Newton method. In this chapter we introduce the conjugate gradient method which is one between the steepest descent method and the Newton method. The conjugate gradient method deflects the direction of the steepest descent method by adding to it a positive multiple of the direction used in the last step. This method only requires the first-order derivatives but overcomes the steepest descent method's shortcoming of slow convergence. At the same time, the method need not save and compute the second-order derivatives which are needed by Newton method. In particular, since it does not require the Hessian matrix or its approximation, it is widely used to solve large scale optimization problems.

In this chapter, we will discuss the derivation, the properties, the algorithm and numerical experiments, and the convergence of the conjugate gradient method. Note that the restarting and preconditioning are very important to improve the conjugate gradient method. As a beginning, we first introduce the concept of conjugate directions and the conjugate direction method.

4.1 Conjugate Direction Methods

One of the main properties of the conjugate gradient method is that its directions are conjugate. Now, we first introduce conjugate directions and conjugate direction methods.

Definition 4.1.1 Let G be an $n \times n$ symmetric and positive definite matrix, $d_1, d_2, \dots, d_m \in R^n$ be non-zero vectors, $m \leq n$. If

$$d_i^T G d_j = 0, \quad \forall i \neq j, \quad (4.1.1)$$

the vectors $d_1, d_2, \dots, d_m$ are called G -conjugate or simply conjugate.

Obviously, if vectors $d_1, \dots, d_m$ are G -conjugate, then they are linearly independent. If $G = I$, the conjugacy is equivalent to the usual orthogonality.

A general conjugate direction method has the following steps:

Algorithm 4.1.2 (*General Conjugate Direction Method*)

Step 1. Given an initial point $x_0, \epsilon > 0, k := 0$. Compute $g_0 = g(x_0)$;
Compute d_0 such that $d_0^T g_0 < 0$.

Step 2. If $\|g_k\| \leq \epsilon$, stop.

Step 3. Compute α_k such that

$$f(x_k + \alpha_k d_k) = \min_{\alpha \geq 0} f(x_k + \alpha d_k).$$

Set $x_{k+1} = x_k + \alpha_k d_k$.

Step 4. Compute d_{k+1} by some conjugate direction method, such that $d_{k+1}^T G d_j = 0, j = 0, 1, \dots, k$.

Step 5. Set $k := k + 1$, go to Step 2. $\square$

The conjugate direction method is an important class of optimization methods. The following theorem shows that, under exact line search, the conjugate direction methods have quadratic termination property, which means that the method terminates in at most n steps when it is applied to a quadratic function with positive definite Hessian.

Theorem 4.1.3 (*Principal Theorem of Conjugate Direction Method*) For a quadratic function with positive definite Hessian G , the conjugate direction method terminates in at most n exact line searches. Each x_{i+1} is the minimizer in the subspace generated by x_0 and the directions $d_0, \dots, d_i$, that is $\{x \mid x = x_0 + \sum_{j=0}^i \alpha_j d_j\}$.

Proof. Since G is positive definite and the conjugate directions $d_0, d_1, \dots$ are linearly independent, it is enough to prove for all $i \leq n-1$ that

$$g_{i+1}^T d_j = 0, \quad j = 0, \dots, i. \quad (4.1.2)$$

(Note that if (4.1.2) holds, we immediately have $g_n^T d_j = 0, j = 0, \dots, n-1$ and $g_n = 0$, therefore x_n is a minimizer.)

To prove (4.1.2), we consider two cases $j < i$ and $j = i$. Keep in mind that

$$y_k \stackrel{Def}{=} g_{k+1} - g_k = G(x_{k+1} - x_k) = \alpha_k G d_k. \quad (4.1.3)$$

When $j < i$, by use of exact line search and the conjugacy, we have

$$\begin{aligned} g_{i+1}^T d_j &= g_{j+1}^T d_j + \sum_{k=j+1}^i y_k^T d_j \\ &= g_{j+1}^T d_j + \sum_{k=j+1}^i \alpha_k d_k^T G d_j \\ &= 0. \end{aligned} \quad (4.1.4)$$

When $j = i$, (4.1.2) is a direct result from the exact line search. Thus (4.1.2) holds and we complete the proof. $\square$

This theorem is simple but important. All conjugate direction methods rely on this theorem. We reemphasize that, under exact line search, all conjugate direction methods satisfy (4.1.2), and have quadratic termination property. This shows that conjugacy plus exact line search implies quadratic termination.

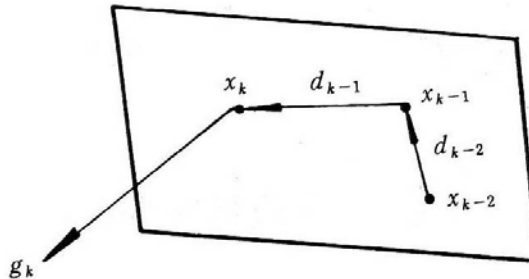


Figure 4.1.1 The gradient of conjugate direction method satisfies (4.1.2)

4.2 Conjugate Gradient Method

4.2.1 Conjugate Gradient Method

In the conjugate direction method described in §4.1, there is not an explicit procedure for generating a conjugate system of vectors $d_1, d_2, \dots$. In this section we describe a method for generating mutually conjugate direction vectors, which is theoretically appealing and computationally effective. This method is called the conjugate gradient method.

In conjugate direction methods, the conjugate gradient method is of particular importance. Now it is widely used to solve large scale optimization problems. The conjugate gradient method was originally proposed by Hestenes and Stiefel [173] in the 1950s to solve linear systems. Since solving a linear system is equivalent to minimizing a positive definite quadratic function, Fletcher and Reeves [138] in the 1960s modified it and developed a conjugate gradient method for unconstrained minimization. By means of conjugacy, the conjugate gradient method makes the steepest descent direction have conjugacy, and thus increases the efficiency and reliability of the algorithm.

Now we derive the conjugate gradient method for the quadratic case.

Let

$$f(x) = \frac{1}{2}x^T Gx + b^T x + c, \quad (4.2.1)$$

where G is an $n \times n$ symmetric positive definite matrix, $b \in R^n$ and c is a real number. Obviously, the gradient of $f(x)$ is

$$g(x) = Gx + b. \quad (4.2.2)$$

Set

$$d_0 = -g_0, \quad (4.2.3)$$

then we have

$$x_1 = x_0 + \alpha_0 d_0, \quad (4.2.4)$$

where α_0 is generated by an exact line search. Then we have

$$g_1^T d_0 = 0. \quad (4.2.5)$$

Set

$$d_1 = -g_1 + \beta_0 d_0, \quad (4.2.6)$$

and choose β_0 such that

$$d_1^T G d_0 = 0. \quad (4.2.7)$$

It follows from multiplying (4.2.6) by $d_0^T G$ that

$$\beta_0 = \frac{g_1^T G d_0}{d_0^T G d_0} = \frac{g_1^T (g_1 - g_0)}{d_0^T (g_1 - g_0)} = \frac{g_1^T g_1}{g_0^T g_0}. \quad (4.2.8)$$

In general, in the k -th iteration, set

$$d_k = -g_k + \sum_{i=0}^{k-1} \beta_i d_i. \quad (4.2.9)$$

Choosing β_i such that $d_k^T G d_i = 0, i = 0, 1, \dots, k-1$, and noticing from Theorem 4.1.3 that

$$g_k^T d_i = 0, g_k^T g_i = 0, i = 0, 1, \dots, k-1, \quad (4.2.10)$$

it follows from multiplying (4.2.9) by $d_j^T G$, ($j = 0, 1, \dots, k-1$) that

$$\beta_j = \frac{g_k^T G d_j}{d_j^T G d_j} = \frac{g_k^T (g_{j+1} - g_j)}{d_j^T (g_{j+1} - g_j)}, j = 0, 1, \dots, k-1. \quad (4.2.11)$$

Then

$$\beta_j = 0, j = 0, 1, \dots, k-2, \quad (4.2.12)$$

$$\beta_{k-1} = \frac{g_k^T (g_k - g_{k-1})}{d_{k-1}^T (g_k - g_{k-1})} = \frac{g_k^T g_k}{g_{k-1}^T g_{k-1}}. \quad (4.2.13)$$

The above derivation establishes the iterative scheme of the conjugate gradient method:

$$x_{k+1} = x_k + \alpha_k d_k, \quad (4.2.14)$$

$$d_k = -g_k + \beta_{k-1} d_{k-1}, \quad (4.2.15)$$

where

$$\beta_{k-1} = \frac{g_k^T g_k}{g_{k-1}^T g_{k-1}}, \text{ (F-R Formula)} \quad (4.2.16)$$

and α_k is an exact step size, in particular, for the quadratic case,

$$\alpha_k = \frac{-g_k^T d_k}{d_k^T G d_k}. \quad (4.2.17)$$

The other famous formulas of β_k are as follows:

$$\beta_{k-1} = \frac{g_k^T(g_k - g_{k-1})}{d_{k-1}^T(g_k - g_{k-1})}, \text{ (H-S or C-W Formula)} \quad (4.2.18)$$

$$\beta_{k-1} = \frac{g_k^T(g_k - g_{k-1})}{g_{k-1}^T g_{k-1}}, \text{ (PRP Formula)} \quad (4.2.19)$$

$$\beta_{k-1} = -\frac{g_k^T g_k}{d_{k-1}^T g_{k-1}}, \text{ (Dixon Formula)} \quad (4.2.20)$$

$$\beta_{k-1} = \frac{g_k^T g_k}{d_{k-1}^T(g_k - g_{k-1})}, \text{ (D-Y Formula)} \quad (4.2.21)$$

where F-R, H-S (or C-W), PRP, Dixon and D-Y formula refer respectively Fletcher-Reeves formula, Hestenes-Stiefel (or Crowder-Wolfe) formula, Polak-Ribière-Polyak formula, Dixon formula and Dai-Yuan Formula. It is easy to see that these formulas are equivalent in the sense that all yield the same search directions when used in minimizing a quadratic function with positive definite Hessian matrix. However, for a general nonlinear function with inexact line search, their behavior is markedly different. Some descriptions will be given later in this subsection.

From (4.2.14)-(4.2.16), we can see that the conjugate gradient method is only a little more complex than the steepest descent method, but it has quadratic termination property and need not compute the Hessian or its approximation matrix. Besides, we will learn below that the conjugate gradient method has global convergence and n -step local quadratic convergence. Hence this method is very attractive especially for large scale optimization problems.

The following theorem includes the main properties of a conjugate gradient method.

Theorem 4.2.1 (*Property theorem of conjugate gradient method*) *For positive definite quadratic function (4.2.1), the conjugate gradient method (4.2.14)-(4.2.16) with exact line searches terminates after $m \leq n$ steps, and the following properties hold for all i , ($0 \leq i \leq m$),*

$$d_i^T G d_j = 0, \quad j = 0, 1, \dots, i-1, \quad (4.2.22)$$

$$g_i^T g_j = 0, \quad j = 0, 1, \dots, i-1, \quad (4.2.23)$$

$$d_i^T g_i = -g_i^T g_i, \quad (4.2.24)$$

$$[g_0, g_1, \dots, g_i] = [g_0, Gg_0, \dots, G^i g_0], \quad (4.2.25)$$

$$[d_0, d_1, \dots, d_i] = [g_0, Gg_0, \dots, G^i g_0]. \quad (4.2.26)$$

where m is the number of distinct eigenvalues of G .

Proof. We prove (4.2.22)–(4.2.24) by induction. For $i = 1$, it is trivial. Suppose (4.2.22)–(4.2.24) hold for some $i < m$. We show that they also hold for $i + 1$.

For quadratic function (4.2.1), we have obviously

$$g_{i+1} = g_i + G(x_{i+1} - x_i) = g_i + \alpha_i Gd_i. \quad (4.2.27)$$

From (4.2.17), α_i can be written as

$$\alpha_i = \frac{g_i^T g_i}{d_i^T G d_i} \neq 0. \quad (4.2.28)$$

Using (4.2.27) and (4.2.15) gives

$$\begin{aligned} g_{i+1}^T g_j &= g_i^T g_j + \alpha_i d_i^T G g_j \\ &= g_i^T g_j - \alpha_i d_i^T G(d_j - \beta_{j-1} d_{j-1}). \end{aligned} \quad (4.2.29)$$

When $j = i$, (4.2.29) becomes

$$g_{i+1}^T g_i = g_i^T g_i - \frac{g_i^T g_i}{d_i^T G d_i} d_i^T G d_i = 0.$$

When $j < i$, (4.2.29) is zero directly by induction hypothesis. So, (4.2.23) follows.

Now, from (4.2.15) and (4.2.27), it follows that

$$\begin{aligned} d_{i+1}^T G d_j &= -g_{i+1}^T G d_j + \beta_i d_i^T G d_j \\ &= g_{i+1}^T (g_j - g_{j+1}) / \alpha_j + \beta_i d_i^T G d_j. \end{aligned} \quad (4.2.30)$$

When $j = i$, it follows from (4.2.30), (4.2.23), (4.2.28) and (4.2.16) that

$$d_{i+1}^T G d_i = -\frac{g_{i+1}^T g_{i+1}}{g_i^T g_i} d_i^T G d_i + \frac{g_{i+1}^T g_{i+1}}{g_i^T g_i} d_i^T G d_i = 0.$$

When $j < i$, (4.2.30) is also zero from induction hypothesis. Then (4.2.22) follows.

Also, from (4.2.15) and the exact line search, we have

$$\begin{aligned} d_{i+1}^T g_{i+1} &= -g_{i+1}^T g_{i+1} + \beta_i d_i^T g_{i+1} \\ &= -g_{i+1}^T g_{i+1}, \end{aligned}$$

which shows (4.2.24) holds for $i + 1$.

Finally, we show (4.2.25) and (4.2.26) by induction. It is trivial for $i = 0$. Now suppose they hold for some i , and we prove that they hold also for $i + 1$.

From the induction hypothesis, both g_i and Gd_i belong to

$$[g_0, Gg_0, \dots, G^i g_0, G^{i+1} g_0].$$

Then it follows from (4.2.27) that $g_{i+1} \in [g_0, Gg_0, \dots, G^{i+1} g_0]$. Furthermore, we need to show

$$g_{i+1} \notin [g_0, Gg_0, \dots, G^i g_0] = [d_0, \dots, d_i].$$

In fact, since vectors $d_0, \dots, d_i$ are conjugate, it follows from Theorem 4.1.3 that $g_{i+1} \perp [d_0, \dots, d_i]$. If $g_{i+1} \in [g_0, Gg_0, \dots, G^i g_0] = [d_0, \dots, d_i]$, then it results in $g_{i+1} = 0$. This is a contradiction. Therefore (4.2.25) follows.

Similarly, by (4.2.15) and induction hypothesis, we can get (4.2.26). $\square$

In this theorem, (4.2.22)–(4.2.24) represent respectively conjugacy of directions, orthogonality of gradients, and descent condition. (4.2.25)–(4.2.26) give some relations between direction vectors and gradients. Usually, The subspace $[g_0, Gg_0, \dots, G^i g_0]$ is called the Krylov subspace.

Recall please the convergence rate (3.1.7), (3.1.8), and (3.1.9) of the steepest descent method for quadratic functions in Theorem 3.1.5. Similarly, for quadratic functions, we can also obtain the following facts for the conjugate gradient method:

Fact 1:

$$\frac{\|x_k - x^*\|_G}{\|x_0 - x^*\|_G} \leq \left(\frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1} \right)^k, \quad (4.2.31)$$

where κ is the spectral condition number of G .

Fact 2: starting from x_1 , the iterate x_{k+2} of the conjugate gradient method after $k + 1$ iterations satisfies

$$E(x_{k+2}) \leq \left(\frac{\lambda_{k+1} - \lambda_n}{\lambda_{k+1} + \lambda_n} \right)^2 E(x_1) = \left(\frac{1 - \lambda_n/\lambda_{k+1}}{1 + \lambda_n/\lambda_{k+1}} \right)^2 E(x_1), \quad (4.2.32)$$

where $E(x)$ is defined by

$$E(x) = \frac{1}{2}(x - x^*)^T G(x - x^*),$$

and the eigenvalues λ_i of G satisfy

$$\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_k \geq \lambda_{k+1} \geq \cdots \geq \lambda_n > 0.$$

Clearly, after the first iteration ($k = 0$), the obtained iterate x_2 satisfies

$$E(x_2) \leq \left(\frac{\lambda_1 - \lambda_n}{\lambda_1 + \lambda_n} \right)^2 E(x_1)$$

which is the same as convergence rate (3.1.8) of the steepest descent method, this is because, at the first iteration, the direction of the conjugate gradient method just is the steepest descent direction. However, after the second iteration ($k = 1$), we have

$$E(x_3) \leq \left(\frac{\lambda_2 - \lambda_n}{\lambda_2 + \lambda_n} \right)^2 E(x_1).$$

At this time, the influence of the largest eigenvalue λ_1 has been removed. The formula (4.2.32) indicates that after each additional iteration of the conjugate gradient method, the influence of one bigger eigenvalue will be removed.

Next, we would like to discuss restart strategy. Since the direction d_k after n steps is no longer conjugate for general non-quadratic functions, it is suitable to reset periodically d_k to the steepest descent direction, i.e., set

$$d_{cn} = -g_{cn}, \quad c = 1, 2, \dots$$

This strategy is called restart. With this strategy, the resultant x_{n-1} is nearer to x^* than x_0 . Especially, when the iterate enters from an area in which non-quadratic behavior is strong into a neighborhood in which a quadratic model function approximates $f(x)$ well, the restart method is able to converge rapidly. For large scale problems, restart strategy will be used more frequently, for example, every k iterations restart, where $k < n$, even $k \ll n$.

Notice that the restart conjugate gradient method permits inexact line search. However, some control measures are needed so that the resultant direction is descending. In fact, we have

$$g_k^T d_k = -g_k^T g_k + \beta_{k-1} g_k^T d_{k-1}. \quad (4.2.33)$$

If exact line search was used in the previous iteration, then $g_k^T d_{k-1} = 0$, and hence $g_k^T d_k = -g_k^T g_k < 0$ which guarantees that d_k is a descent direction. However, if inexact line search was used in the previous iteration, the quantity $\beta_{k-1} g_k^T d_{k-1}$ may be positive and larger than $-g_k^T g_k$, consequently $-g_k^T g_k + \beta_{k-1} g_k^T d_{k-1}$ is possibly larger than zero. In this case d_k will not be a descent direction. A typical remedy for such an eventuality is to restart the algorithm with d_k as the steepest descent direction $-g_k$. However, frequently setting d_k to the steepest descent direction will lessen the efficiency of the algorithm, and make the behavior of the algorithm incline to a steepest descent method. This situation requires care. The following control measure can be used to overcome this difficulty.

Let $\bar{g}_{k+1}$, $\bar{d}_{k+1}$ and $\bar{\beta}_k$ denote the computed values of g_{k+1} , d_{k+1} and β_k at $x_k + \alpha^j d_k$ respectively, where $\{\alpha^j\}$ is a test step size sequence generated from a step size algorithm. If

$$-\bar{g}_{k+1}^T \bar{d}_{k+1} \geq \sigma \|\bar{g}_{k+1}\|_2 \|\bar{d}_{k+1}\|_2, \quad (4.2.34)$$

where σ is a small positive number, then α^j is accepted as α_k . If (4.2.34) is not satisfied at any trial points, we will use exact line search to produce α_k .

The following algorithm is a restart conjugate gradient method with exact line search.

Algorithm 4.2.2 (*Restart F-R Conjugate Gradient Method*)

Step 0. Given x_0 , $\epsilon > 0$.

Step 1. Set $k = 0$. Compute $g_0 = g(x_0)$.

Step 2. If $\|g_0\| \leq \epsilon$, stop; otherwise, set $d_0 = -g_0$.

Step 3. Compute step size α_k , such that

$$f(x_k + \alpha_k d_k) = \min_{\alpha \geq 0} \{f(x_k + \alpha d_k)\}.$$

Step 4. Set $x_{k+1} = x_k + \alpha_k d_k$, $k := k + 1$.

Step 5. Compute $g_k = g(x_k)$. If $\|g_k\| \leq \epsilon$, stop; otherwise go to Step 6.

Step 6. If $k = n$, set $x_0 = x_k$, and go to Step 1; otherwise, go to Step 7.

Step 7. Compute $\beta = g_k^T g_k / g_{k-1}^T g_{k-1}$, $d_k = -g_k + \beta d_{k-1}$.

Step 8. If $d_k^T g_k > 0$, set $x_0 = x_k$, and go to Step 1; otherwise go to Step 3. $\square$

(4.2.18)–(4.2.20) are common formulas of the conjugate gradient method. The Fletcher-Reeves formula (4.2.16) is the first presented in 1964 for solving optimization problems and now is the most widely used in practice. However, in general, this formula does not have the descent property and is often used in conjunction with exact line search. Dixon's formula (4.2.20) has descent property. If we employ inexact line search

$$|g_{k+1}^T d_k| \leq -\sigma g_k^T d_k, \quad 0 < \sigma < 1,$$

Dixon's formula satisfies

$$d_k^T g_k < 0, \text{ if } g_k \neq 0.$$

The Polak-Ribiere-Polyak (PRP) formula (4.2.19) has a characteristic that it can restart automatically. When the algorithm goes slowly and $g_{k+1} \approx g_k$, PRP formula will produce $\beta_k \approx 0$ and thus $d_{k+1} \approx -g_{k+1}$. This indicates that the algorithm has a tendency of restarting automatically, so that it can overcome some shortcomings of going forward slowly. Various numerical experiments show that PRP formula is more robust and efficient than other existing formulas for solving optimization problems.

4.2.2 Beale's Three-Term Conjugate Gradient Method

Beale [10] considered the three-term conjugate gradient method. The idea is as follows. When frequently periodic restarts with the steepest descent direction are used, the reduction at the restart iteration is often poor compared with the reduction that would have occurred without restarting. However, if the restart direction is taken as an arbitrary vector, the required conjugacy relations may not hold. Now we consider restarting at x_t , and take the direction d_t generated by the algorithm as the restarting direction to begin the new cycle, and require the constructed sequence of directions to satisfy the conjugacy.

Set

$$d_{t+1} = -g_{t+1} + \beta_t d_t, \quad (4.2.35)$$

$$d_k = -g_k + \gamma_{k-1} d_t + \beta_{t+1} d_{t+1} + \cdots + \beta_{k-1} d_{k-1}, \quad (4.2.36)$$

where $n + t - 1 \geq k \geq t + 2$. Similar to the derivation of the traditional conjugate gradient method, by means of conjugacy between d_{t+1} and d_t , d_k and d_t , $d_{t+1}, \dots, d_{k-1}$, we can get the following relation:

$$\beta_{k-1} = \frac{g_k^T G d_{k-1}}{d_{k-1}^T G d_{k-1}}, \quad \gamma_{k-1} = \frac{g_k^T G d_t}{d_t^T G d_t},$$

$$\beta_j = 0, \quad j = t + 1, \dots, k - 2.$$

Then (4.2.36) can be reduced as

$$d_k = -g_k + \beta_{k-1} d_{k-1} + \gamma_{k-1} d_t, \quad (4.2.37)$$

where

$$\beta_{k-1} = \frac{g_k^T (g_k - g_{k-1})}{d_{k-1}^T (g_k - g_{k-1})}, \quad (4.2.38)$$

$$\gamma_{k-1} = \begin{cases} 0, & \text{if } k = t + 1; \\ \frac{g_k^T (g_{t+1} - g_t)}{d_t^T (g_{t+1} - g_t)}, & \text{if } k > t + 1. \end{cases} \quad (4.2.39)$$

Note that β_{k-1} in (4.2.38) can be represented as any formula in (4.2.18)-(4.2.21), for example,

$$\beta_{k-1} = \frac{g_k^T g_k}{g_{k-1}^T g_{k-1}}$$

which is F-R formula.

Note also that in Beale's three-term formula (4.2.37), d_k may not be a descent direction, even if exact line searches are made. In order to make d_k be sufficient downhill and make two consecutive gradients not be far from orthogonal, we may impose some control measures as follows,

$$-g_k^T d_k \geq \sigma \|g_k\| \|d_k\|, \quad (4.2.40)$$

where σ is a small positive number, and

$$|g_{k-1}^T g_k| < 0.2 \|g_k\|^2. \quad (4.2.41)$$

Since the iterate x_k generated from (4.2.37)-(4.2.39) is a minimizer of the linear manifold

$$\begin{aligned} \mathcal{B}_{k-1} &= x_t + [d_t, d_{t+1}, \dots, d_{k-1}] \\ &= x_t + [d_t, g_{t+1}, \dots, g_{k-1}], \end{aligned}$$

and hence

$$g_k \perp [d_t, d_{t+1}, \dots, d_{k-1}]$$

and

$$g_k \perp [d_t, g_{t+1}, \dots, g_{k-1}].$$

Below, we give Beale's three-term conjugate gradient algorithm.

Algorithm 4.2.3 (*Beale's three-term CG method*)

Step 1. Given x_0 , set $k = 0, t = 0$, evaluate $g_0 = g(x_0)$. If $\|g_0\| \leq \epsilon$, stop; otherwise set $d_0 = -g_0$.

Step 2. Compute α_k by exact line search.

Step 3. Set $x_{k+1} = x_k + \alpha_k d_k$, set $k := k + 1$, evaluate $g_k = g(x_k)$.

Step 4. If $\|g_k\| \leq \epsilon$, stop; otherwise go to Step 5.

Step 5. If both conditions

$$|g_{k-1}^T g_k| \geq 0.2 \|g_k\|^2$$

and

$$k - t \geq n - 1$$

do not hold, go to Step 7; otherwise go to Step 6.

Step 6. Set $t = k - 1$.

Step 7. Compute d_k by (4.2.37)-(4.2.39).

Step 8. If $k > t + 1$, go to Step 9; otherwise go to Step 2.

Step 9 If

$$-1.2 \|g_k\|^2 \leq d_k^T g_k \leq -0.8 \|g_k\|^2,$$

go to Step 2; otherwise go to Step 6. $\square$

4.2.3 Preconditioned Conjugate Gradient Method

In the discussion above we have known that if the conjugate gradient method is applied to minimize the quadratic function

$$f(x) = \frac{1}{2}x^T Gx + b^T x + c, \quad (4.2.42)$$

where G is symmetric and positive definite, it computes the solution of the system

$$Gx = -b. \quad (4.2.43)$$

In this case, the algorithm is called the linear conjugate gradient method, and the notation r is used for the gradient vector $Gx_k + b$, which, in fact, is the residual of the system (4.2.43).

The linear conjugate gradient method is as follows: given x_0 and $r_0 = Gx_0 + b$, $\beta_{-1} = 0$, $d_{-1} = 0$, and each iteration includes the following steps for $k = 0, 1, \dots$:

$$\begin{aligned} d_k &= -r_k + \beta_{k-1}d_{k-1}, \\ \alpha_k &= \frac{r_k^T r_k}{d_k^T G d_k}, \\ x_{k+1} &= x_k + \alpha_k d_k, \\ r_{k+1} &= r_k + \alpha_k G d_k, \\ \beta_k &= \frac{r_{k+1}^T r_{k+1}}{r_k^T r_k}. \end{aligned} \quad (4.2.44)$$

If exact arithmetic is used, the convergence of the linear conjugate gradient method will be achieved in $m(\leq n)$ iterations, where m is the number of distinct eigenvalues of G . If the eigenvalues of G are clustered into groups of approximately equal value, the method may converge very quickly. However, for general eigenvalue structure, due to rounding errors, considerably more than n iterations may be required. Hence, the convergence rate depends on the structure of eigenvalues of G and the condition number of G . If the original system is replaced by an equivalent system in which the conditioning of G is improved, then the convergence rate can be improved. This technique is called preconditioning.

Consider the transformation

$$x = C^{-1}z,$$

where C is a nonsingular matrix. The solution of $Gx = -b$ is equivalent to solving the linear system

$$C^{-T}GC^{-1}z = -C^{-T}b.$$

If we adequately choose C such that the condition number of $C^{-T}GC^{-1}$ is as small as possible, the convergence rate of the algorithm will be improved. Since $C^{-T}GC^{-1}$ is similar to $W^{-1}G$, where $W = C^TC$, it means that we should choose W such that the condition number of $W^{-1}G$ is as small as possible.

The preconditioned conjugate gradient method is as follows: given x_0 , set $g_0 = Gx_0 + b$, and let $v_0 = W^{-1}g_0$ and $d_0 = -v_0$. For $k = 0, 1, \dots$,

$$\alpha_k = \frac{g_k^T v_k}{d_k^T G d_k}, \quad (4.2.45)$$

$$x_{k+1} = x_k + \alpha_k d_k, \quad (4.2.46)$$

$$g_{k+1} = g_k + \alpha_k G d_k, \quad (4.2.47)$$

$$v_{k+1} = W^{-1}g_{k+1}, \quad (4.2.48)$$

$$\beta_k = \frac{g_{k+1}^T v_{k+1}}{g_k^T v_k}, \quad (4.2.49)$$

$$d_{k+1} = -v_{k+1} + \beta_k d_k. \quad (4.2.50)$$

The preconditioning matrix W can be defined in several ways. The simplest strategy is to choose W as the diagonal of G . In this case, the condition number of $W^{-1}G$ is bounded by $(1 + \delta)/(1 - \delta)$, where $\delta \ll 1$. The popular strategy for preconditioning is use of incomplete Cholesky factorization. The basic idea is as follows. Instead of computing the exact Cholesky factor L which satisfies $G = LL^T$, we compute an approximate factor $\tilde{L}$ which is more sparse than L , such that $G \approx \tilde{L}\tilde{L}^T$, and then choose $C = \tilde{L}^T$, and hence $W = \tilde{L}\tilde{L}^T$,

$$C^{-T}GC^{-1} = \tilde{L}^{-1}G\tilde{L}^{-T} \approx I$$

and

$$W^{-1}G = (\tilde{L}\tilde{L}^T)^{-1}G \approx I.$$

In this procedure, any fill-in during the sparse Cholesky factorization is discarded.

The other preconditioning matrix can be obtained by performing a limited-memory quasi-Newton method. From the quasi-Newton method (see Chapter 5) the limited memory matrix M satisfies the quasi-Newton condition for

r ($r \ll n$) pairs of vectors $\{s_j, y_j\}$,

$$s_j = My_j, \quad j = 1, \dots, r,$$

where $s_j = x_{j+1} - x_j$, $y_j = g_{j+1} - g_j$. Since $Gs_j = y_j$, we have

$$s_j = MGs_j,$$

and the matrix MG has r unit eigenvalues with eigenvectors $\{s_j\}$. Therefore, M can be used as W^{-1} .

For the minimization of a non-quadratic function, the preconditioning matrix W is varied from iteration to iteration. In this case, we consider

$$x = C^{-1}z, \quad (4.2.51)$$

and the objective function is transformed as

$$f(x) = f(C^{-1}z) = \tilde{f}(z). \quad (4.2.52)$$

Set

$$z_k = Cx_k, \quad \tilde{g}_k = \nabla \tilde{f}(z_k) = C^{-T} \nabla f(x_k) = C^{-T} g_k,$$

then

$$\tilde{d}_k = Cd_k, \quad \tilde{s}_k = Cs_k, \quad \tilde{y}_k = C^{-T} y_k.$$

So, application of conjugate gradient method, for example (4.2.18), to $\tilde{f}(z)$ yields the direction

$$\begin{aligned} \tilde{d}_{k+1} &= -\tilde{g}_{k+1} + \frac{\tilde{g}_{k+1}^T(\tilde{g}_{k+1} - \tilde{g}_k)}{\tilde{d}_k^T(\tilde{g}_{k+1} - \tilde{g}_k)} \tilde{d}_k \\ &= -\left(I - \frac{\tilde{d}_k \tilde{y}_k^T}{\tilde{d}_k^T \tilde{y}_k}\right) \tilde{g}_{k+1}, \end{aligned} \quad (4.2.53)$$

and hence

$$\begin{aligned} d_{k+1} &= -\left(I - \frac{d_k y_k^T}{d_k^T y_k}\right) W^{-1} g_{k+1}, \\ &\triangleq -P_{k+1} g_{k+1} \end{aligned} \quad (4.2.54)$$

which is the formula of the preconditioned conjugate gradient method, where $W = C^T C$. Similarly, we can obtain

$$d_{k+1} = -\left[I - \frac{1}{y_k^T s_k} (y_k s_k^T + s_k y_k^T) + \left(1 + \frac{y_k^T y_k}{y_k^T s_k}\right) \frac{s_k s_k^T}{y_k^T s_k}\right] W^{-1} g_{k+1} \quad (4.2.55)$$

which is the preconditioned conjugate gradient method in BFGS formula without memory.

In general, the preconditioning matrix is varied with different problems. There is not a general-purpose formula for preconditioners.

4.3 Convergence of Conjugate Gradient Methods

As for the convergence results of the conjugate gradient method for minimizing a general non-quadratic function, there have been various results. In this section, we introduce global convergence results of conjugate gradient methods due to Zoutendijk [385], Polyak [255] and Al-Baali [2] etc., and also give in brief the outline of local convergence rates obtained by Cohen [61], and McCormick and Ritter [205].

4.3.1 Global Convergence of Conjugate Gradient Methods

This subsection is divided into two parts. The first part discusses the global convergence of conjugate gradient methods with exact line search, and consists of three theorems which state respectively global convergence of Fletcher-Reeves (F-R) conjugate gradient method, Crowder-Wolfe (C-W) conjugate gradient method, and Polak-Ribière-Polyak (PRP) conjugate gradient method. The second part discusses the global convergence of F-R conjugate gradient method with inexact line search.

Now, we start the discussion by proving the global convergence result of F-R method in the case of exact line search.

Theorem 4.3.1 (*Global convergence of F-R conjugate gradient method*)

Suppose that $f : R^n \rightarrow R$ is continuously differentiable on a bounded level set $L = \{x \in R^n \mid f(x) \leq f(x_0)\}$, and that F-R conjugate gradient method is implemented with exact line search. Then the produced sequence $\{x_k\}$ has at least one accumulation point which is a stationary point, i.e.,

(1) when $\{x_k\}$ is a finite sequence, then the final point x^ is a stationary point of f ;*

(2) when $\{x_k\}$ is an infinite sequence, it has limit point, and any limit point is a stationary point.

Proof. (1) When $\{x_k\}$ is finite, from the termination condition, it follows that the final point x^* satisfies $\nabla f(x^*) = 0$, and hence x^* is a stationary point of f .

(2) When $\{x_k\}$ is infinite, we have $\nabla f(x_k) \neq 0, \forall k$. Noting that $d_k = -g_k + \beta_{k-1}d_{k-1}$ and $g_k^T d_{k-1} = 0$ by exact line search, we have

$$g_k^T d_k = -\|g_k\|^2 + \beta_{k-1}g_k^T d_{k-1} = -\|g_k\|^2 < 0, \quad (4.3.1)$$

which means that d_k is a descent direction, $\{f(x_k)\}$ is a monotone descent sequence, and thus $\{x_k\} \subset L$. Therefore $\{x_k\}$ is a bounded sequence and must have a limit point.

Let x^* be a limit point of $\{x_k\}$. Then there is a subsequence $\{x_k\}_{K_1}$ converging to x^* , where K_1 is an index set of a subsequence of $\{x_k\}$. Since $\{x_k\}_{K_1} \subset \{x_k\}$, $\{f(x_k)\}_{K_1} \subset \{f(x_k)\}$. It follows from the continuity of f that for $k \in K_1$,

$$f(x^*) = f(\lim_{k \rightarrow \infty} x_k) = \lim_{k \rightarrow \infty} f(x_k) = f^*. \quad (4.3.2)$$

Similarly, $\{x_{k+1}\}$ is also a bounded sequence. Hence there exists a subsequence $\{x_{k+1}\}_{K_2}$ converging to $\bar{x}^*$, where K_2 is an index set of a subsequence of $\{x_{k+1}\}$. In this case,

$$f(\bar{x}^*) = f(\lim_{k \rightarrow \infty} x_{k+1}) = \lim_{k \rightarrow \infty} f(x_{k+1}) = f^*. \quad (4.3.3)$$

Then

$$f(\bar{x}^*) = f(x^*) = f^*. \quad (4.3.4)$$

Now we prove $\nabla f(x^*) = 0$ by contradiction. Suppose that $\nabla f(x^*) \neq 0$, then, for α sufficiently small, we have

$$f(x^* + \alpha d^*) < f(x^*). \quad (4.3.5)$$

Since

$$f(x_{k+1}) = f(x_k + \alpha_k d_k) \leq f(x_k + \alpha d_k), \quad \forall \alpha > 0,$$

then for $k \in K_2$, passing to limit $k \rightarrow \infty$ and using (4.3.5), we get

$$f(\bar{x}^*) \leq f(x^* + \alpha d^*) < f(x^*), \quad (4.3.6)$$

which contradicts (4.3.4). This proves $\nabla f(x^*) = 0$, i.e., x^* is a stationary point of f . $\square$

Similarly, we can state the global convergence of Crowder-Wolfe (C-W) restart conjugate gradient method with exact line search as follows.

Theorem 4.3.2 (*Global convergence of Crowder-Wolfe conjugate gradient method*) Suppose that the level set $L = \{x \in R^n \mid f(x) \leq f(x_0)\}$ is bounded, and that $\nabla f(x)$ is Lipschitz continuous. Assume that Crowder-Wolfe conjugate gradient method is implemented with exact line search and restart strategy. Then the produced sequence $\{x_k\}$ has at least one accumulation point which is a stationary point.

Proof. See Polyak [255]. $\square$

As mentioned before, PRP method is more efficient than F-R method. We naturally hope PRP method has also the above property for a general non-quadratic function. Unfortunately, the above Theorem 4.3.1 is not true for PRP method (see Powell [270]). However, with stronger condition that f is uniformly convex, the PRP method is globally convergent. The following theorem states this result.

Theorem 4.3.3 Let $f(x)$ be twice continuously differentiable and the level set $L = \{x \in R^n \mid f(x) \leq f(x_0)\}$ be bounded. Suppose that there is a constant $m > 0$ such that for $x \in L$,

$$m\|y\|^2 \leq y^T \nabla^2 f(x) y, \quad \forall y \in R^n. \quad (4.3.7)$$

Then the sequence $\{x_k\}$ generated by PRP method with exact line search converges to the unique minimizer x^* of f .

Proof. From Theorem 2.2.4, we know that it is enough to prove that (2.2.13) holds, that is, there exists a constant $\rho > 0$ such that

$$-g_k^T d_k \geq \rho \|g_k\| \|d_k\|, \quad (4.3.8)$$

which means

$$\cos \theta_k \geq \rho > 0.$$

Then, from Theorem 2.2.4, we have $g_k \rightarrow 0$ and $g(x^*) = 0$. From (4.3.7), it follows that $\{x_k\} \rightarrow x^*$ which is a unique minimizer.

By using $g_k^T d_{k-1} = 0$ and (4.2.15), we have

$$g_k^T d_k = -\|g_k\|^2.$$

Then (4.3.8) is equivalent to

$$\frac{\|g_k\|}{\|d_k\|} \geq \rho. \quad (4.3.9)$$

From (4.2.17) and (4.2.15), it follows that

$$\alpha_{k-1} = -\frac{g_{k-1}^T d_{k-1}}{d_{k-1}^T G_{k-1} d_{k-1}} = \frac{\|g_{k-1}\|^2}{d_{k-1}^T G_{k-1} d_{k-1}}, \quad (4.3.10)$$

where

$$G_{k-1} = \int_0^1 G(x_{k-1} + t\alpha_{k-1}d_{k-1})dt. \quad (4.3.11)$$

By (4.3.11), the integral form of the mean-value theorem is

$$g_k - g_{k-1} = g(x_{k-1} + \alpha_{k-1}d_{k-1}) - g(x_{k-1}) = \alpha_{k-1}G_{k-1}d_{k-1}. \quad (4.3.12)$$

Then, by (4.3.11) and (4.3.10), (4.2.19) becomes

$$\begin{aligned} \beta_{k-1} &= \frac{g_k^T (g_k - g_{k-1})}{g_{k-1}^T g_{k-1}} = \alpha_{k-1} \frac{g_k^T G_{k-1} d_{k-1}}{\|g_{k-1}\|^2} \\ &= \frac{g_k^T G_{k-1} d_{k-1}}{d_{k-1}^T G_{k-1} d_{k-1}}. \end{aligned} \quad (4.3.13)$$

Since the level set L is bounded, there is a constant $M > 0$, such that

$$y^T G(x)y \leq M\|y\|^2, \quad x \in L, \forall y \in R^n. \quad (4.3.14)$$

Then, by (4.3.13), (4.3.14) and (4.3.7), we have

$$|\beta_{k-1}| \leq \frac{\|g_k\| \|G_{k-1} d_{k-1}\|}{m \|d_{k-1}\|^2} \leq \frac{M}{m} \frac{\|g_k\|}{\|d_{k-1}\|}. \quad (4.3.15)$$

Therefore

$$\begin{aligned} \|d_k\| &\leq \|g_k\| + |\beta_{k-1}| \|d_{k-1}\| \\ &\leq \|g_k\| + \frac{M}{m} \|g_k\| \\ &= \left(1 + \frac{M}{m}\right) \|g_k\|, \end{aligned} \quad (4.3.16)$$

which gives

$$\frac{\|g_k\|}{\|d_k\|} \geq \left(1 + \frac{M}{m}\right)^{-1}. \quad (4.3.17)$$

The above inequality shows that (4.3.9) holds. We complete the proof. $\square$

Next, we discuss the case of inexact line search. Al-Baali [2] studied the F-R conjugate gradient method with strong Wolfe-Powell rule (2.5.3) and (2.5.9), and proved the global convergence. The following theorem indicates that, in the inexact case, the search direction d_k satisfies descent property: $g_k^T d_k < 0$.

Theorem 4.3.4 *If, for all k , α_k are determined by strong Wolfe-Powell rule (2.5.3) and (2.5.9), then for F-R-CG method, the inequality*

$$-\sum_{j=0}^k \sigma^j \leq \frac{g_k^T d_k}{\|g_k\|^2} \leq -2 + \sum_{j=0}^k \sigma^j \quad (4.3.18)$$

holds for all k , and hence the descent property

$$g_k^T d_k < 0, \quad \forall k \quad (4.3.19)$$

holds, as long as $g_k \neq 0$.

Proof. The proof is by induction. For $k = 0$, $d_0 = -g_0$, $\sigma^0 = 1$, hence (4.3.18) and (4.3.19) hold for $k = 0$.

Now we suppose that (4.3.18) and (4.3.19) hold for any $k \geq 0$. By (4.2.15) and (4.2.16), we have

$$\frac{g_{k+1}^T d_{k+1}}{\|g_{k+1}\|^2} = -1 + \frac{g_{k+1}^T d_k}{\|g_k\|^2}. \quad (4.3.20)$$

Using (2.5.9) and induction assumption (4.3.19) yields

$$-1 + \sigma \frac{g_k^T d_k}{\|g_k\|^2} \leq \frac{g_{k+1}^T d_{k+1}}{\|g_{k+1}\|^2} \leq -1 - \sigma \frac{g_k^T d_k}{\|g_k\|^2}. \quad (4.3.21)$$

Also, by induction assumption (4.3.18), we have

$$\begin{aligned} -\sum_{j=0}^{k+1} \sigma^j &= -1 - \sigma \sum_{j=0}^k \sigma^j \leq \frac{g_{k+1}^T d_{k+1}}{\|g_{k+1}\|^2} \\ &\leq -1 + \sigma \sum_{j=0}^k \sigma^j = -2 + \sum_{j=0}^{k+1} \sigma^j. \end{aligned}$$

Then, (4.3.18) holds for $k + 1$.

Since

$$\frac{g_{k+1}^T d_{k+1}}{\|g_{k+1}\|^2} \leq -2 + \sum_{j=0}^{k+1} \sigma^j \quad (4.3.22)$$

and

$$\sum_{j=0}^{k+1} \sigma^j < \sum_{j=0}^{\infty} \sigma^j = \frac{1}{1-\sigma}, \quad (4.3.23)$$

where $\sigma \in (0, 1)$, it follows from $1 - \sigma > \frac{1}{2}$ that $-2 + \sum_{j=0}^{k+1} \sigma^j < 0$. Hence, from (4.3.22), we obtain $g_{k+1}^T d_{k+1} < 0$. We complete the proof by induction. $\square$

Now, we are in a position to prove the global convergence of F-R-CG algorithm with inexact line search.

Theorem 4.3.5 *Let f be twice continuously differentiable, and the level set $L = \{x \in R^n \mid f(x) \leq f(x_0)\}$ be bounded. Suppose that the steplength α_k is determined by strong Wolfe-Powell rule (2.5.3) and (2.5.9), where $0 < \rho < \sigma < \frac{1}{2}$. Then the sequence $\{x_k\}$ generated by F-R-CG method is globally convergent, i.e.,*

$$\liminf_{k \rightarrow \infty} \|g_k\| = 0. \quad (4.3.24)$$

Proof. By (2.5.9), (4.3.18) and (4.3.23), we have

$$|g_k^T d_{k-1}| \leq -\sigma g_{k-1}^T d_{k-1} \leq \sigma \sum_{j=0}^{k-1} \sigma^j \|g_{k-1}\|^2 \leq \frac{\sigma}{1-\sigma} \|g_{k-1}\|^2. \quad (4.3.25)$$

Also, by (4.2.15), (4.3.25) and (4.2.16), we obtain

$$\begin{aligned} \|d_k\|^2 &= \|g_k\|^2 - 2\beta_{k-1} g_k^T d_{k-1} + \beta_{k-1}^2 \|d_{k-1}\|^2 \\ &\leq \|g_k\|^2 + \frac{2\sigma}{1-\sigma} \|g_k\|^2 + \beta_{k-1}^2 \|d_{k-1}\|^2 \\ &= \left(\frac{1+\sigma}{1-\sigma} \right) \|g_k\|^2 + \beta_{k-1}^2 \|d_{k-1}\|^2. \end{aligned} \quad (4.3.26)$$

By applying this relation repeatedly, it follows that

$$\|d_k\|^2 \leq \left(\frac{1+\sigma}{1-\sigma} \right) \|g_k\|^4 \left(\sum_{j=0}^k \|g_j\|^{-2} \right), \quad (4.3.27)$$

where we used the facts that

$$\beta_k^2 \beta_{k-1}^2 \cdots \beta_{k-i}^2 = \frac{\|g_k\|^2}{\|g_{k-i-1}\|^2}.$$

Now we prove (4.3.24) by contradiction. It assumes that (4.3.24) does not hold, then there exists a constant $\epsilon > 0$ such that

$$\|g_k\| \geq \epsilon > 0 \quad (4.3.28)$$

holds for all k sufficiently large. Since g_k is bounded above on the level set L , it follows from (4.3.27) that

$$\|d_k\|^2 \leq c_1 k, \quad (4.3.29)$$

where c_1 is a positive constant. From (4.3.18) and (4.3.23), we have

$$\begin{aligned} \cos \theta_k &= -\frac{g_k^T d_k}{\|g_k\| \|d_k\|} \geq \left(2 - \sum_{j=0}^k \sigma^j\right) \frac{\|g_k\|}{\|d_k\|} \\ &\geq \left(\frac{1-2\sigma}{1-\sigma}\right) \frac{\|g_k\|}{\|d_k\|}. \end{aligned} \quad (4.3.30)$$

Since $\sigma < \frac{1}{2}$, substituting (4.3.29) and (4.3.28) into (4.3.30) gives

$$\sum_k \cos^2 \theta_k \geq \left(\frac{1-2\sigma}{1-\sigma}\right)^2 \sum_k \frac{\|g_k\|^2}{\|d_k\|^2} \geq c_2 \sum_k \frac{1}{k}, \quad (4.3.31)$$

where c_2 is a positive constant. Therefore, the series $\sum_k \cos^2 \theta_k$ is divergent.

Let M be an upper bound of $\|G(x)\|$ on the level set L , then

$$g_{k+1}^T d_k = (g_k + \alpha_k G(x_k) d_k)^T d_k \leq g_k^T d_k + \alpha_k M \|d_k\|^2.$$

By using (2.5.9), i.e., $\sigma g_k^T d_k \leq g_{k+1}^T d_k \leq -\sigma g_k^T d_k$, we obtain

$$\alpha_k \geq -\frac{1-\sigma}{M \|d_k\|^2} g_k^T d_k. \quad (4.3.32)$$

Substituting α_k of (4.3.32) into (2.5.3) gives

$$\begin{aligned} f_{k+1} &\leq f_k - \frac{(1-\sigma)\rho}{M} \left(\frac{g_k^T d_k}{\|d_k\|}\right)^2 \\ &= f_k - c_3 \|g_k\|^2 \cos^2 \theta_k, \end{aligned}$$

where $c_3 = \frac{(1-\sigma)\rho}{M} > 0$. Since $f(x)$ is bounded below, $\sum_k \|g_k\|^2 \cos^2 \theta_k$ converges, which indicates that $\sum_k \cos^2 \theta_k$ converges by use of (4.3.28). This fact contradicts (4.3.31). We complete the proof. $\square$

In the above theorem, the conclusion is also true if, instead of f being twice continuously differentiable, the assumptions on f are changed: let f be continuously differentiable and bounded below, and ∇f be Lipschitz continuous.

To conclude the subsection, we give the global convergence of D-Y conjugate gradient method with Wolfe-Powell rule.

Theorem 4.3.6 *Let x_1 be a starting point, $f(x)$ be continuously differentiable and bounded below on the level set L , $\nabla f(x)$ satisfy the Lipschitz condition on L . Let α_k satisfy Wolfe-Powell rule (2.5.3) and (2.5.7). Then, for all k ,*

$$g_k^T d_k < 0,$$

and further

$$\liminf_{k \rightarrow \infty} \|g_k\| = 0.$$

Proof. See Dai and Yuan [75]. $\square$

4.3.2 Convergence Rate of Conjugate Gradient Methods

We have already seen that the conjugate gradient method has quadratic termination, that is, for a convex quadratic function, the conjugate gradient method with exact line search terminates after n iterations.

In (4.2.31) and (4.2.32), we give two formulas for convergence rate of conjugate gradient method, from which we have seen that, for a quadratic function, the rate of convergence of conjugate gradient methods is not worse than that of the steepest descent method; that is, it is not worse than linear. Furthermore, we can also have the following demonstration. For convenience, we assume

$$f(x) = \frac{1}{2} x^T G x, \quad (4.3.33)$$

where G is an $n \times n$ positive definite matrix. Clearly, the explicit expression of steplength is

$$\alpha_k = -\frac{d_k^T G x_k}{d_k^T G d_k} = -\frac{d_k^T g_k}{d_k^T G d_k}. \quad (4.3.34)$$

So, we can obtain

$$\begin{aligned}
 f(x_{k+1}) &= \frac{1}{2} x_{k+1}^T G x_{k+1} \\
 &= \frac{1}{2} (x_k + \alpha_k d_k)^T G (x_k + \alpha_k d_k) \\
 &= \frac{1}{2} x_k^T G x_k - \frac{1}{2} \frac{(g_k^T d_k)^2}{d_k^T G d_k}.
 \end{aligned} \tag{4.3.35}$$

In the case of the steepest descent (SD) method we have $d_k = -g_k$ and thus

$$f(x_{SD}^{k+1}) = \frac{1}{2} x_k^T G x_k - \frac{1}{2} \frac{\|g_k\|^4}{g_k^T G g_k}. \tag{4.3.36}$$

Whereas in the case of the conjugate gradient (CG) method, we have $d_k = -g_k + \beta_{k-1} d_{k-1}$ and thus

$$f(x_{CG}^{k+1}) = \frac{1}{2} x_k^T G x_k - \frac{1}{2} \frac{\|g_k\|^4}{d_k^T G d_k} \tag{4.3.37}$$

$$= f(x_k) - \frac{1}{2} \frac{\|g_k\|^4}{d_k^T G d_k}. \tag{4.3.38}$$

Since

$$\begin{aligned}
 d_k^T G d_k &= (-g_k + \beta_{k-1} d_{k-1})^T G (-g_k + \beta_{k-1} d_{k-1}) \\
 &= g_k^T G g_k + \beta_{k-1}^2 d_{k-1}^T G d_{k-1} \\
 &\leq g_k^T G g_k,
 \end{aligned}$$

it follows that

$$f(x_{CG}^{k+1}) \leq f(x_{SD}^{k+1}). \tag{4.3.39}$$

The above discussion indicates again that the conjugate gradient method reduces the value of f at least as much as the steepest descent method. Since the steepest descent method has a linear convergence rate, we conclude that conjugate gradient methods have convergence rates that are no worse than the linear rate. From (4.3.38) we also know that, for conjugate gradient methods, the objective value is strictly decreasing. Similarly, the result is true for the preconditioned conjugate gradient method and we have

$$f(x_{CG}^{k+1}) = f(x_k) - \frac{1}{2} \frac{(g_k^T v_k)^2}{d_k^T G d_k}, \tag{4.3.40}$$

where $v_k = W^{-1}g_k$.

Note that the conjugate gradient method with exact line search can find the minimizer of a convex quadratic function in at most n iterations, which corresponds to one step of Newton method. Hence we can say that if n iterations of the conjugate gradient method are regarded as a big iteration, the conjugate gradient method should have a similar convergence rate as Newton method. Cohen [61], Burmeister [39], and McCormick and Ritter [205] studied the n -step quadratic convergence rate. We now state this result without proof in the following theorem.

Assume that

(A1) $f : R^n \rightarrow R$ is three times continuously differentiable;

(A2) there exist constants $M > m > 0$ such that

$$m\|y\|^2 \leq y^T \nabla^2 f(x)y \leq M\|y\|^2, \quad \forall y \in R^n, x \in L, \quad (4.3.41)$$

where L is a bounded level set.

Theorem 4.3.7 *Assume that the conditions (A1) and (A2) are satisfied, then the sequence $\{x_k\}$ generated by PRP-CG and F-R-CG restart methods have n -step quadratic convergence rate, that is, there exists a constant $c > 0$, such that*

$$\limsup_{k \rightarrow \infty} \frac{\|x_{kr+n} - x^*\|}{\|x_{kr} - x^*\|^2} \leq c < \infty, \quad (4.3.42)$$

where r means that the methods restart per r iterations.

Further, Ritter [287] shows that the convergence rate is n -step super-quadratic, that is,

$$\|x_{k+n} - x^*\| = o(\|x_k - x^*\|^2). \quad (4.3.43)$$

The other results on convergence rate of conjugate gradient methods can consult Stoer [325].

Exercises

1. Let G be an $n \times n$ symmetric positive definite matrix, $p_1, p_2, \dots, p_n$ be n linearly independent vectors. Define

$$\begin{aligned} d_1 &= p_1, \\ d_{k+1} &= p_{k+1} - \sum_{i=1}^k \frac{p_{k+1}^T G d_i}{d_i^T G d_i} d_i, \quad k = 1, 2, \dots, n-1. \end{aligned}$$

Prove that $\{d_k\}$ are G -conjugate.

2. Using F-R conjugate gradient method minimize the following functions:

(1) $f(x) = x_1^2 + 2x_2^2 - 2x_1x_2 + 2x_2 + 2$, the initial point $x^{(0)} = (0, 0)^T$.

(2) $f(x) = (x_1 - 1)^4 + (x_1 - x_2)^2$, the initial point $x^{(0)} = (0, 0)^T$.

3. Using respectively F-R conjugate gradient method and PRP conjugate gradient method minimize the Rosenbrock function in Appendix 1.1 and Extended Rosenbrock function in Appendix 1.2.

4. Try to prove respectively that $\{d_k\}$ generated by PRP-CG method and Dixon-CG method are conjugate.

5. Derive the Beale three-term conjugate gradient formula (4.2.38)–(4.2.39).

6. Let $f(x) = \frac{1}{2}x^T Ax - b^T x$, where A is an $n \times n$ symmetric positive definite matrix. Setting $x_{k+1} = x_k + \alpha_k d_k$ and $d_k = -r_k + \beta_k d_{k-1}$, prove

(1) the exact step size $\alpha_k = -\frac{r_k^T d_k}{d_k^T A d_k}$,

(2) $\beta_k = \frac{r_k^T A d_{k-1}}{d_{k-1}^T A d_{k-1}}$.

7. Using the linear conjugate gradient method minimize function $f(x) = \frac{1}{2}x^T Ax - b^T x$, where A is a Hilbert matrix $A = \left(\frac{1}{i+j-1}\right)$, $b = (1, 1, \dots, 1)^T$, the initial point $x^{(0)} = 0$. Try considering the cases of $n = 5, 10, 20$.